



Inflammatory markers associated with dementia: a systematic review and meta-analysis

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ABSTRACT

Background: Dementia has been linked to several modifiable risk factors, including mental illness. The inflammatory-mediated neurodegeneration hypothesis posits a causal relationship between the two whereby mental illness triggers an inflammatory response, which in turn acts as a catalyst for the neurodegenerative changes that lead to dementia. Existing meta-analyses have yet to investigate inflammatory markers in Alzheimer's disease (AD), vascular dementia (VaD), dementia with Lewy bodies (DLB), or fronto-temporal dementia (FTD) in the context of the inflammatory-mediated neurodegeneration hypothesis. This meta-analysis therefore explored whether a) AD, VaD, DLB, and FTD are each associated with greater inflammation than healthy controls, independent of comorbid mental or physical health problems with a known inflammatory response, and b) there are any similarities in the inflammatory profiles associated with these dementia subtypes.

Methods: PubMed, EMBASE, PsycINFO and CINAHL searches identified 41 eligible studies.

Results: AD is associated with an inflammatory response, with tentative evidence to indicate that VaD, DLB, and FTD are also associated with inflammation. However, the inflammatory response appears to differ across these conditions.

Limitations: The data for VaD, DLB, FTD and a number of inflammatory markers were limited.

Conclusions: Although tentative, AD, VaD, DLB, and FTD appear to be associated with discrete inflammatory processes that are not attributable to other common comorbid mental or physical health problems that cause inflammation. Whether this inflammatory response plays a causal role in the development of dementia and/or is triggered by prior mental illness remains to be determined.

1. Introduction

Dementia is recognised as a public health priority, with over 55 million people worldwide living with dementia; a figure that is projected to increase to more than 130 million by 2050 (World Health Organization, 2017). Alzheimer's disease (AD), vascular dementia (VaD), dementia with Lewy bodies (DLB) and fronto-temporal dementia (FTD) are the most common subtypes, together accounting for over 90% of dementia diagnoses in people aged 65 years and older (Alzheimer's Disease International et al., 2012). In 2019, the estimated annual global care, treatment and lost productivity costs associated with dementia exceeded US\$1.3 trillion and are expected to surpass US\$2.8 trillion by 2030 (World Health Organization, 2021). However, up to one third of dementia cases may be preventable or delayed through active treatment of modifiable risk factors (Livingston et al., 2017). Even a 10% reduction

in modifiable risk factors would prevent an estimated 3 million cases of AD by 2050 (Barnes and Yaffe, 2011; Norton et al., 2014).

Growing evidence from multiple systematic reviews and meta-analyses suggests that mental illness may be one of these modifiable risk factors. More specifically, a history of depression (Cherbuin et al., 2015; Diniz et al., 2013; Ford et al., 2018; Kuring et al., 2020; Xu et al., 2015), anxiety (Becker et al., 2018; Ford et al., 2018; Gulpers et al., 2016; Kuring et al., 2020; Santabarbara et al., 2019) and, possibly, PTSD (Rafferty et al., 2017) have each been linked to an increased risk of later developing dementia. However, it remains uncertain whether these mental illnesses are *prodromal* symptoms and/or are *causally* implicated in the development of dementia.

If they are prodromal, then treatment of these mental health conditions should improve the quality of life of individuals but not reduce their risk of dementia. Alternatively, if causally related, then successful

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treatment of these conditions may reduce the incidence of dementia, resulting in significant economic and health benefits. One mechanism proposed to provide a causal link between mental illness and the development of dementia is via ‘inflammatory-mediated neurodegeneration’ (Leonard, 2006), whereby mental illness is thought to cause neuroinflammatory reactions that trigger neurodegenerative changes; potentially leading to dementia (Leonard, 2017). In theory, if mental illness is a causal risk factor for dementia mediated by inflammatory processes, then people with dementia should exhibit neuroinflammatory changes that are not apparent in healthy persons.

Inflammation is the immune system's way of protecting the body against harm from injury, infection and foreign bodies or substances (Ward, 2010). A protective immune response can also be triggered by chronic stress, which is often associated with mental illness (Leonard, 2006). Immune responses within both the central (CNS) and peripheral (PNS) nervous systems can trigger inflammatory mediators and some immune cells to cross the blood-brain barrier because of bidirectional communication between the CNS and PNS during immune surveillance (Marchetti and Engelhardt, 2020). In the context of disease, however, these changes can alter blood-brain barrier function and promote neuroinflammation (Angiulli et al., 2021). Blood samples are often used as a proxy for CNS measures of neuroinflammation because they provide a reliable measure of neuroinflammatory mediators in both the PNS and CNS (Angiulli et al., 2021; de la Monte et al., 2023; Malpetti et al., 2025; Sabiron et al., 2023; Weinhofer et al., 2025). However, their reliability can differ depending on the specific mediator and study population that is being examined, with brain imaging and cerebrospinal fluid (CSF) samples considered to provide more direct measures of CNS inflammation (Gigase et al., 2023). Inflammation occurs when the immune cells that act as inflammatory sensors (e.g. lymphocytes, monocytes, neutrophils) generate or activate various molecules that mediate this response (e.g. cytokines, leukotrienes) (Ward, 2010). Thus, if the neurodegenerative changes seen in dementia are the result of aberrant inflammation, then marker levels of both immune cells and inflammatory mediators should differ between people with dementia and their healthy peers.

Consistent with this, a number of meta-analyses have identified inflammatory changes in persons with AD (Anuradha et al., 2022; Bradburn et al., 2019; Chen et al., 2018; Darweesh et al., 2018; Gautam et al., 2023; Hecheng et al., 2023; Holper et al., 2024; Jiang et al., 2018; Lai et al., 2017; Liu et al., 2018; Mavroudis et al., 2021; Ng et al., 2018; Shen et al., 2019; Su et al., 2019; Swardfager et al., 2010; Zhou et al., 2023a) and all-cause/unspecified dementia (Darweesh et al., 2018; Gibson et al., 2021; Koyama et al., 2013; Pan et al., 2024), when compared to their healthy peers. Critically, however, these studies were not specifically designed to examine the ‘inflammatory-mediated neurodegenerative’ hypothesis and therefore have several limitations when reviewing their results in this context. First, none of them excluded studies that did not screen for comorbid mental (e.g. depression, anxiety, PTSD) and physical health conditions (e.g. arthritis, diabetes), making it unclear whether the inflammation was associated with dementia or other comorbid health conditions that have a known inflammatory response (e.g. autoimmune or infectious disorders). Second, most studies examined a limited number of inflammatory markers (Anuradha et al., 2022; Bradburn et al., 2019; Chen et al., 2018; Darweesh et al., 2018; Gautam et al., 2023; Holper et al., 2024; Jiang et al., 2018; Koyama et al., 2013; Liu et al., 2018; Mavroudis et al., 2021; Ng et al., 2018; Shen et al., 2019) or one type of inflammatory mediator (e.g. cytokines) (Gibson et al., 2021; Hecheng et al., 2023; Swardfager et al., 2010; Zhou et al., 2023a). A more inclusive examination of inflammatory markers, including both immune cells (e.g. lymphocytes, microglia) and inflammatory mediators (e.g. cytokines, leukotrienes), is required to further our understanding of the role of inflammation in the development of dementia. Third, most meta-analyses only examined AD, with very few including VaD, DLB or FTD (e.g. Darweesh et al., 2018; Gibson et al., 2021) and none provided separate data for these subtypes; instead categorising them as ‘all-cause’

dementia. A comparison of the inflammatory profiles associated with the most common dementia subtypes is important to improve our understanding of the relationship between inflammation and the different dementias. Finally, some of these meta-analyses are quite dated and, consequently, do not include recent research in this rapidly-growing area (Koyama et al., 2013; Swardfager et al., 2010).

The current meta-analysis was designed to examine the ‘inflammatory mediated neurodegenerative’ hypothesis by determining whether there is evidence of inflammation associated with dementia after excluding co-morbid, inflammatory related conditions. Specifically, the aims were to: a) examine whether AD, VaD, DLB, and FTD are associated with greater inflammation than their healthy peers, independent of comorbid mental or physical health problems that have a known inflammatory response, and b) explore whether there are any similarities in the inflammatory profiles associated with these dementia subtypes. Unlike existing meta-analyses, the current study: (1) only examined studies that screened for, and excluded, common comorbid mental (e.g. depression) and physical health conditions (e.g. autoimmune disorders) with a known inflammatory response; (2) included all immune cells and the molecules they produce (intracellular, extracellular [secreted], or expressed on the cell surface) to capture a broad range of immune cells and inflammatory mediators; and (3) investigated each of the four most common dementia subtypes (AD/VaD/DLB/FTD), excluding any mixed dementias. Consistent with most earlier meta-analyses, it: (1) excluded studies that examined genes because the presence of a gene does not inevitably result in the expression of the associated inflammatory protein; and (2) only included studies that used published diagnostic criteria to diagnose dementia to ensure that the findings were not confounded by the inclusion of persons with other types of cognitive dysfunction (e.g. delirium, acquired brain injury).

2. Method

This research conforms to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Moher et al., 2009) and the Meta-analysis of Observational Studies in Epidemiology (MOOSE) (Stroup et al., 2000) guidelines. The PROSPERO registration number for this study is CRD42020188244.

2.1. Search strategy

The CINAHL, EMBASE, PsycINFO, and PubMed electronic databases were each searched, guided by an expert research librarian, using various terms for dementia and inflammatory markers (see Online Resource A for detailed search terms). The focus for inflammatory markers was on immune cells and the immune mediator molecules that they produce (including both central [e.g. CSF samples, PET imaging] and peripheral [e.g. blood samples] measures). The search was limited to publications from 1980 onwards because this was the year the Diagnostic and Statistical Manual – third edition (DSM-III) (American Psychiatric Association, 1980) was updated to reflect current criteria for dementia and Alzheimer's disease. The final search was conducted on October 1, 2024. The bibliographies of all included studies, published reviews, systematic reviews/meta-analyses and conference abstracts were also examined to identify any additional relevant studies that were not captured by the database searches. All references were managed in EndNote 20.

2.2. Selection criteria

Studies were required to meet the following selection criteria: 1) dementia was diagnosed using published diagnostic criteria (e.g. DSM); 2) participants were adults (18+ years); 3) participants were screened for, and excluded, if found to have comorbid psychiatric disorders and medical conditions with a known inflammatory response (e.g. arthritis, diabetes, chronic kidney/liver/pulmonary disease, cardiac disease,

recent infections, taking major modifiers of the immune system such as corticosteroids); 4) a healthy control group was recruited; 5) data that would enable calculation of the Standardised Mean Difference (SMD) between the inflammatory marker levels of patient and control groups were available; and 6) the research was original, peer-reviewed and published in English.

Studies were excluded if they: 1) reported a case study or case series, or 2) examined levels of artificially-stimulated inflammatory markers because the current focus was on basal immune activity. Where multiple publications examined the same or overlapping samples (i.e. non-independent data), data was extracted solely from the paper that used the most stringent diagnostic criteria or, where equal, the paper with the largest sample size or, where equal, the most recent publication. When both central (e.g. cerebrospinal fluid) and peripheral (e.g. blood) measures of an inflammatory marker were collected for the same sample, both measures were analysed and the results compared. In three cases, the data from multiple publications with overlapping samples were used to maximise the sample size, with the data for each inflammatory marker taken from the study with the largest sample for that marker (Alvarez et al., 2007 + Anton Alvarez et al., 1996 + Cacabelos et al., 1991; Blennow et al., 1990 + Blennow et al., 1991; Hampel et al., 1998 + Hampel et al., 1999).

The first author (JKK) screened all studies by title and abstract and then examined the full-texts of all potentially relevant studies in more detail (see Fig. 1). Where there was any uncertainty, study eligibility was discussed with the other authors (JLM, LW, GT) until a consensus was obtained. The authors of ten recent papers (published within the last 10 years) were emailed to request additional data to facilitate calculation of the SMD. One author responded and the study was included (Magalhaes et al., 2018). The remaining authors were unable to provide the required data or did not respond, necessitating their exclusion (Bednarska-Makaruk et al., 2017; Boccardi et al., 2019; Dursun et al.,

2015; Faria et al., 2014; Janelidze et al., 2018; Mehta et al., 2023; Pawlowski et al., 2018; Wang et al., 2014; Zhang et al., 2021).

2.3. Data extraction and quality assessment

The following data was extracted (JKK) from all studies: (1) study characteristics (author, publication year, sample size, recruitment setting, geographical location), (2) participant demographics (age, sex, ethnicity, education, socioeconomic status, cognitive screening test score), (3) dementia details (type [AD/VaD/DLB/FTD], criteria used to diagnose dementia, duration), (4) details of the inflammatory marker(s) (marker name, peripheral or central measurement), and (5) data to calculate the SMD between the levels of inflammatory markers in participants with dementia and their healthy controls.

An adaptation of the National Institute of Health (NIH) Quality Assessment Tool for Observational and Cohort Cross-Sectional Studies (National Institute of Health, 2014) was used to assess the reporting quality of included studies. Studies were rated (JKK) on whether they met (2 points), partially met (1 point), did not meet (0 points), or did not report (0 points) each of the 11 criteria (see Fig. 2). The same rater (JKK) blindly re-scored a random sample of 21 studies 6 to 8 weeks later to assess the intra-rater reliability of these scores. An intra-class correlation (ICC) measured the consistency between the total scores from both ratings for each study (Rankin and Stokes, 1998), and a percentage score measured the agreement between the first and second ratings assigned to each of the 11 criteria.

2.4. Effect size and statistical analysis

Summary descriptive statistics were calculated using Microsoft Excel, the ICC statistics were calculated using SPSS (version 27), and all remaining analyses were calculated using Comprehensive Meta-Analysis

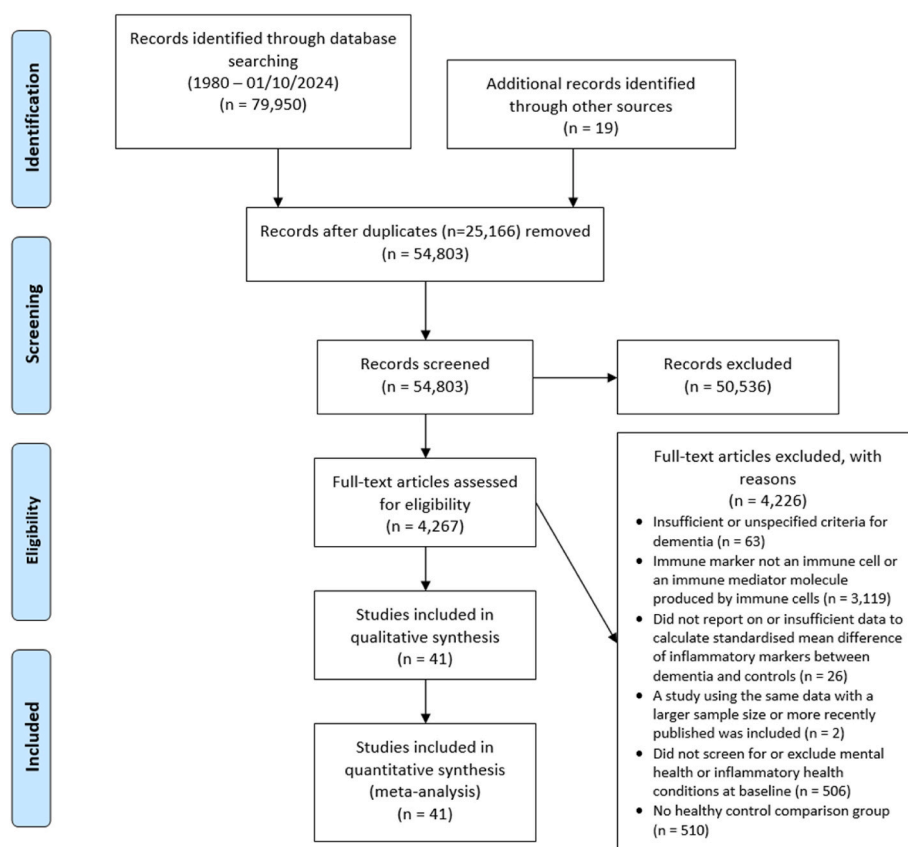


Fig. 1. PRISMA flow chart documenting the literature search and screening processes.

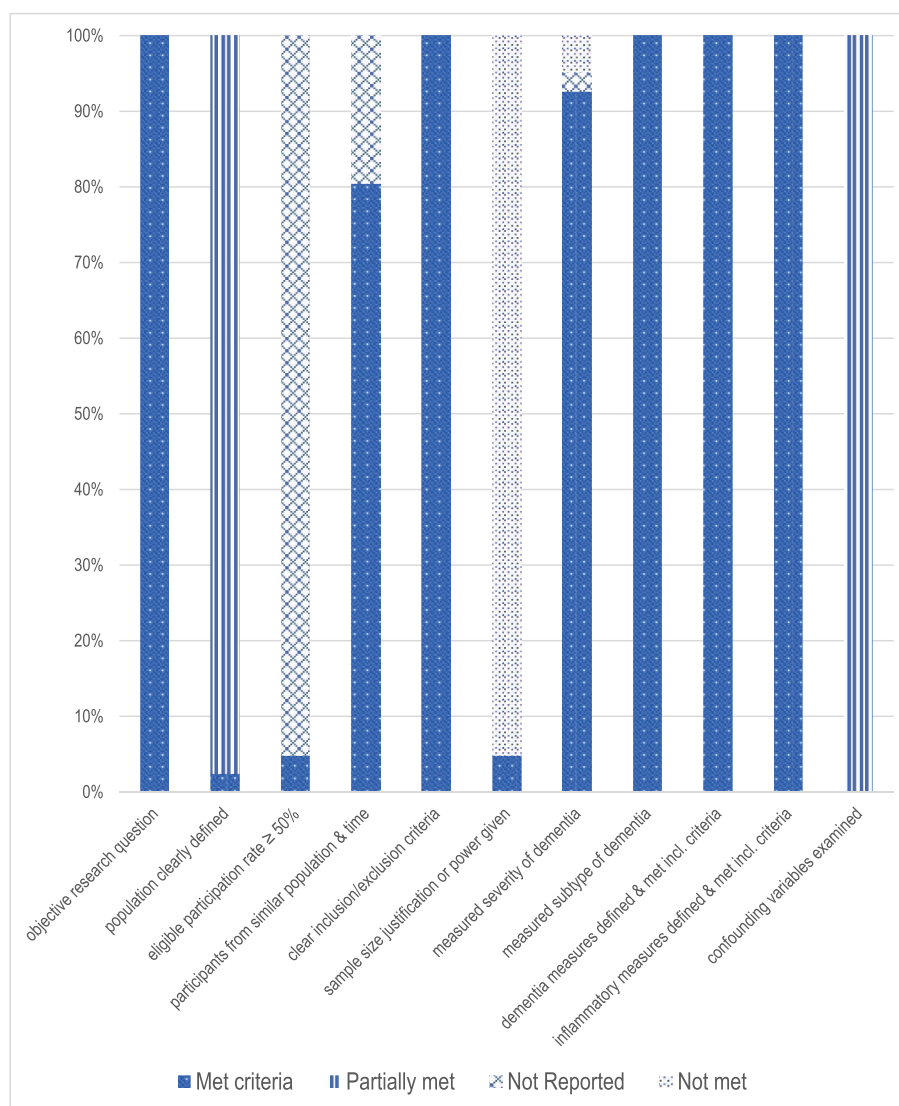


Fig. 2. Percentage of studies meeting each of the adapted NIH study quality criteria.

(CMA; Version 4).

The SMD, in the form of Hedges' g (and 95% Confidence Intervals [CIs]), was used to evaluate the difference between the levels of inflammatory markers in people with dementia (AD/VaD/DLB/FTD) and their healthy controls. *Individual* Hedges' g values were calculated for all inflammatory markers reported by the included studies and *mean* Hedges' g values were calculated for inflammatory markers that were examined by multiple studies (random-effects model, with each study weighted by the inverse of its variance). Whereas *positive* Hedges' g values denote *higher* levels of a marker in people with dementia compared with their controls, *negative* values indicate *lower* levels. An effect size of 1.3 or more is considered very large, 0.8 large, 0.5 medium, and 0.2 small (Sullivan and Feinn, 2012).

Between-study variability in the Hedges' g values for inflammatory markers that were examined by three or more studies was assessed using two statistics: Q (tests for significant heterogeneity) and I^2 (measures the magnitude of heterogeneity). Significant Q values indicate that the differences between the findings from individual studies were unlikely to be due solely to chance, and that a random-effects model should be used to account for any possible clinical (e.g. duration of dementia) or methodological (e.g. participants' age) study differences. I^2 values range from 0 (indicating that the heterogeneity in effect sizes from individual studies is likely due entirely to chance/sampling error) to 100

(indicating that all the variability is due to clinical or methodological study differences). Tau-squared (τ^2) was used to calculate the 95% prediction intervals (i.e. the range within which we expect to find the true effect/Hedges' g).

An examination of moderator (i.e., clinical or methodological) variables via subgroup analysis and meta-regression was planned where there were high levels of heterogeneity or large prediction intervals and a sufficient number of studies ($N \geq 10$ studies per marker) in order to explain some of the between-study heterogeneity in individual Hedges' g values. Data permitting, an analysis of publication bias (to measure the potential impact of unpublished research or studies not captured by the current searches) was undertaken using Duval and Tweedie's (1998) trim-and-fill method. Unfortunately, as will be seen, there was insufficient data to conduct any of these analyses (see sections 3.3, 3.4, 3.5 and 3.6).

3. Results

3.1. Study screening and summary details

As seen in Fig. 1, the literature searches identified 79,950 references, with 19 additional references found in the bibliographies of review articles and included studies. After eliminating 25,166 duplicates, the

remaining 54,803 references were screened initially by title and abstract against the eligibility criteria; 4267 of which underwent full-text review. A further 4226 articles were excluded in this round (see Fig. 1 for the reasons for exclusion), leaving 41 studies that were meta-analysed.

Key demographic information for these 41 studies is summarised in Table 1, with Online Resource B providing the specific details for each

study (see Online Resource C for a list of these references). Most studies ($N_{\text{studies}} = 35$) had sample sizes below 100 in their dementia groups, with the majority ($N_{\text{studies}} = 29$) having fewer than 50 participants. Participants were largely aged between 65 and 75 years (mean = 70.1 years, SD = 4.43) and, where reported, had some high school education. Study samples comprised slightly more females than males. Dementia

Table 1

Summary demographic and study information for the dementia (AD, VaD, DLB, or FTD) and control groups.

	All Participants				Dementia				Controls			
	N_{studies}	$N_{\text{participants}}$	M	SD	N_{studies}	$N_{\text{participants}}$	M	SD	N_{studies}	$N_{\text{participants}}$	M	SD
Sample size	41	3895	95.0	75.8	41	2256	55.0	42.2	41	1639	40.0	36.6
Age (at baseline)	39	3820	70.1	4.4	39	2218	71.4	4.5	39	1602	69.1	4.0
Education (years)	13	1416	10.6	3.1	13	784	10.2	3.2	13	632	11.9	2.3
MMSE					29	1590	19.1	3.2	26	1245	28.6	1.9
Duration of dementia (years)					8	372	3.9	1.2				
			% ^{studies} ^d	% ^{particip.} ^e			% ^{studies} ^d	% ^{particip.} ^e			% ^{studies} ^d	% ^{particip.} ^e
Sex: Female	37	2399(f)	90.2%	65.6%								
Geographical location of data collection	41	3895	100%	100%	41	2256	100%	100%	41	1639	100%	100%
Australia/New Zealand ^a	0	0	0%	0%	0	0	0%	0%	0	0	0%	0%
North America/Canada	8	373	19.5%	9.6%	8	220	19.5%	9.8%	8	153	19.5%	9.3%
South America	1	183	2.4%	4.7%	1	33	2.4%	1.5%	1	150	2.4%	9.2%
Eastern Europe	0	0	0%	0%	0	0	0%	0%	0	0	0%	0%
Western Europe (incl. UK)	20	2106	48.8%	54.1%	20	1253	48.8%	55.5%	20	853	48.8%	52.0%
Asia	10	1206	24.4%	31.0%	10	660	24.4%	29.3%	10	546	24.4%	33.3%
Africa	2	135	4.9%	3.5%	2	90	4.9%	4.0%	2	45	4.9%	2.8%
	N_{studies}	$N_{\text{participants}}$	% ^{studies} ^d	% ^{particip.} ^e	N_{studies}	$N_{\text{participants}}$	% ^{studies} ^d	% ^{particip.} ^e	N_{studies}	$N_{\text{participants}}$	% ^{studies} ^d	% ^{particip.} ^e
Dementia	41	3895	100%	100%	41	2256	100%	100%	41	1639	100%	100%
Alzheimer's disease (AD) ^b	38	3447	92.7%	88.5%	38	1953	92.7%	86.6%	38	1494	92.7%	91.2%
Vascular dementia (VaD)	2	336	4.9%	8.6%	2	102	4.9%	4.5%	2	234	4.9%	14.3%
Dementia with Lewy Bodies (DLB)	3	318	7.3%	8.2%	3	87	7.3%	3.9%	3	231	7.3%	14.1%
Fronto-temporal dementia (FTD)	2	347	4.9%	8.9%	2	114	4.9%	5.1%	2	233	4.9%	14.2%
Inflammatory Markers (multiple studies) ^c												
TNF-alpha	13	1168	31.7%	30.0%	13	692	31.7%	30.7%	13	476	31.7%	29.0%
IL6	11	592	26.8%	15.2%	11	317	26.8%	14.1%	11	275	26.8%	16.8%
IL1beta	8	343	19.5%	8.8%	8	182	19.5%	8.1%	8	161	19.5%	9.8%
IL8 (CXCL8)	6	665	14.6%	17.1%	6	351	14.6%	15.6%	6	314	14.6%	19.2%
IL17A (IL17)	5	233	12.2%	6.0%	5	116	12.2%	5.1%	5	117	12.2%	7.1%
IL10	5	208	12.2%	5.3%	5	104	12.2%	4.6%	5	104	12.2%	6.3%
IFNgamma	4	133	9.8%	3.4%	4	71	9.8%	3.1%	4	62	9.8%	3.8%
IgG	4	644	9.8%	16.5%	4	376	9.8%	16.7%	4	268	9.8%	16.4%
IL1alpha	4	654	9.8%	16.8%	4	374	9.8%	16.6%	4	280	9.8%	17.1%
lymphocytes	4	466	9.8%	12.0%	4	256	9.8%	11.3%	4	210	9.8%	12.8%
GM-CSF	3	147	7.3%	3.8%	3	74	7.3%	3.3%	3	73	7.3%	4.5%
IL12 (IL12p70)	3	142	7.3%	3.6%	3	67	7.3%	3.0%	3	75	7.3%	4.6%
IL18	3	283	7.3%	7.3%	3	160	7.3%	7.1%	3	123	7.3%	7.5%
IL2	3	112	7.3%	2.9%	3	61	7.3%	2.7%	3	51	7.3%	3.1%
CCL2 (MCP-1)	2	226	4.9%	5.8%	2	116	4.9%	5.1%	2	110	4.9%	6.7%
CD3+CD4+	2	170	4.9%	4.4%	2	105	4.9%	4.7%	2	65	4.9%	4.0%
G-CSF	2	473	4.9%	12.1%	2	259	4.9%	11.5%	2	214	4.9%	13.1%
IgM	2	463	4.9%	11.9%	2	256	4.9%	11.3%	2	207	4.9%	12.6%
IL3	2	473	4.9%	12.1%	2	259	4.9%	11.5%	2	214	4.9%	13.1%
M-CSF1	2	438	4.9%	11.2%	2	238	4.9%	10.5%	2	200	4.9%	12.2%
sIL6R	2	102	4.9%	2.6%	2	61	4.9%	2.7%	2	41	4.9%	2.5%
sTNF-R1 (sTNF-alpha _r)	2	121	4.9%	3.1%	2	60	4.9%	2.7%	2	61	4.9%	3.7%
White blood cells	2	240	4.9%	6.2%	2	139	4.9%	6.2%	2	101	4.9%	6.2%

M = mean. SD = standard deviation. MMSE = Mini mental status examination. ^a Two studies were conducted in both North America/Canada and Western Europe (incl. UK). This data has been summarised here under North America/Canada ^b One study reported on AD, VaD, DLB and FTD. One study reported on AD and DLB. The remaining studies reported on a single dementia type. ^c Data only summarised for inflammatory markers examined by 2 or more studies. ^d Percentage based on total number of studies. Only two studies examined more than one dementia type; most reported on multiple inflammatory markers. ^e Percentage based on total participant numbers ($n = 2391$ for dementia, $n = 1722$ for controls).

was typically of moderate severity, as indicated by the mean MMSE score ($N_{\text{studies}} = 29$) (Perneczky et al., 2006). Only 8 studies reported the mean duration of illness, with an average of 3.9 years ($SD = 1.2$). Very few or no studies provided information relating to ethnicity ($N_{\text{studies}} = 3$) and socioeconomic status ($N_{\text{studies}} = 0$), hence they are not reported here. All studies were prospective, most recruited from healthcare settings ($N_{\text{studies}} = 35$; $N_{\text{studies}} = 6$ did not report the recruitment setting) and very few collected central (CNS) measures of inflammatory markers ($N_{\text{studies}} = 9$).

As seen in Table 1, most studies investigated AD ($N_{\text{studies}} = 38$), with only three examining DLB and two each for VaD and FTD (one study examined both AD and DLB in separate samples, and one examined AD, VaD, DLB and FTD). TNF-alpha was the most frequently assessed marker ($N_{\text{studies}} = 13$), followed by interleukin 6 (IL6) ($N_{\text{studies}} = 11$). Only four

other markers were examined by five or more studies, and just 17 additional markers were examined by at least two studies. Fig. 3 shows the 82 inflammatory markers that were examined by the 41 included studies in one or more types of dementia (46 had significant findings, shown in bold). Most markers ($N_{\text{markers}} = 56$) were examined in only one dementia type (see Fig. 3: cells 1, 3, 7 & 15), with the majority investigated in AD ($N_{\text{markers}} = 36$; see Fig. 3: cells 1, 2, 4, 8, 9, & 12).

3.2. Study reporting quality

As shown in Fig. 2, most studies met or partially met nine of the 11 criteria from the adapted NIH Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies (National Institute of Health, 2014). One criterion, study power, was largely unmet; however, this is

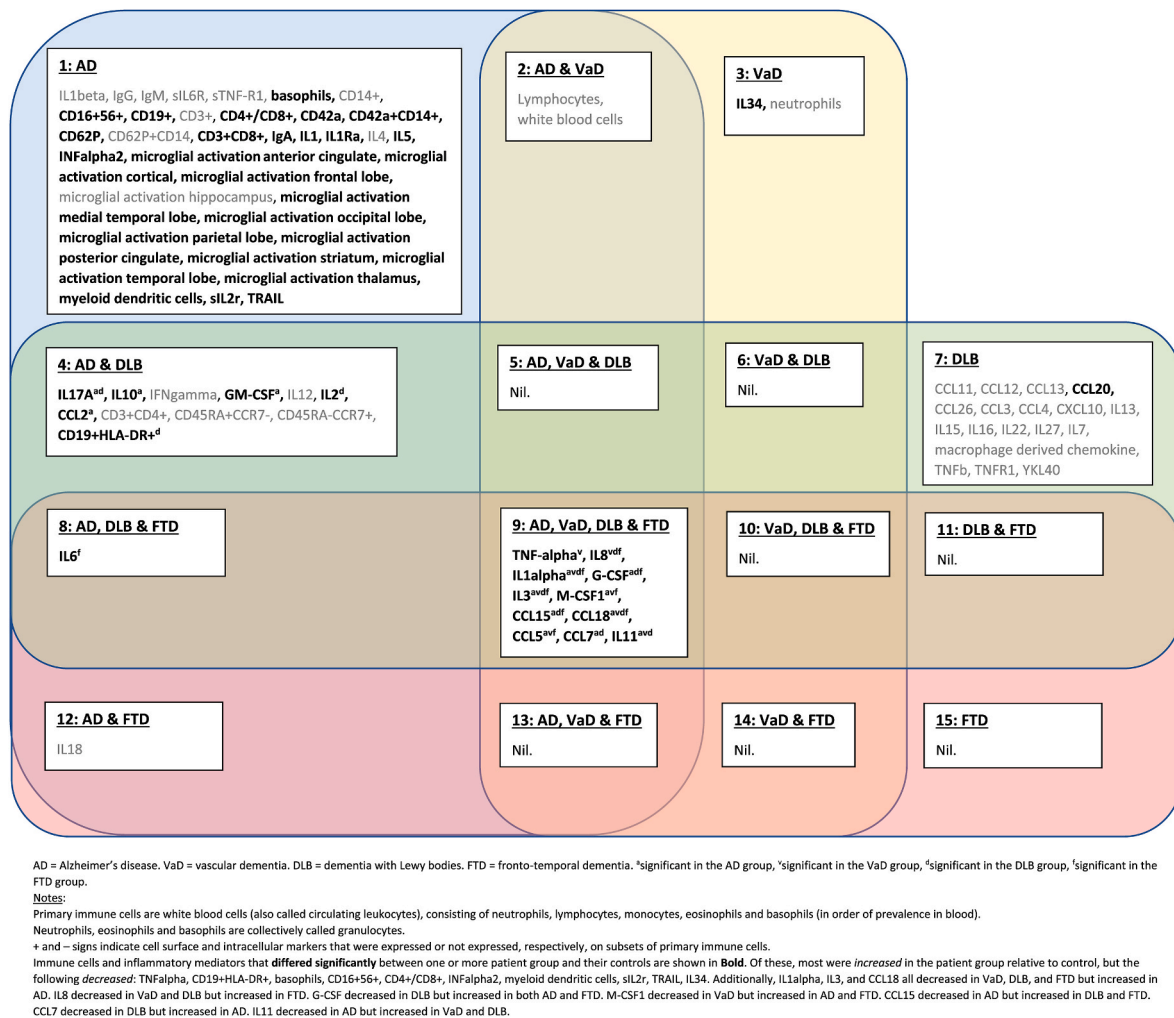


Fig. 3. Immune cells and inflammatory mediators examined in the context of one or more types of dementia [Alzheimer's disease (AD), vascular dementia (VaD), Dementia with Lewy bodies (DLB), or Fronto-temporal dementia (FTD)], listed in order of highest number of studies then alphabetically. AD = Alzheimer's disease. VaD = vascular dementia. DLB = dementia with Lewy bodies. FTD = fronto-temporal dementia. ^asignificant in the AD group, ^vsignificant in the VaD group, ^dsignificant in the DLB group, ^fsignificant in the FTD group.

Notes:
Primary immune cells are white blood cells (also called circulating leukocytes), consisting of neutrophils, lymphocytes, monocytes, eosinophils and basophils (in order of prevalence in blood).
Neutrophils, eosinophils and basophils are collectively called granulocytes.

+ and - signs indicate cell surface and intracellular markers that were expressed or not expressed, respectively, on subsets of primary immune cells.
Immune cells and inflammatory mediators that **differed significantly** between one or more patient group and their controls are shown in **Bold**. Of these, most were *increased* in the patient group relative to control, but the following *decreased*: TNFalpha, CD19+HLA-DR+, basophils, CD16+56+, CD4+/CD8+, INFalpha2, myeloid dendritic cells, sIL2r, TRAIL, IL34. Additionally, IL1alpha, IL3, and CCL18 all decreased in VaD, DLB, and FTD but increased in AD. IL8 decreased in VaD and DLB but increased in FTD. G-CSF decreased in DLB but increased in both AD and FTD. M-CSF1 decreased in VaD but increased in AD and FTD. CCL15 decreased in AD but increased in DLB and FTD. CCL7 decreased in DLB but increased in AD. IL11 decreased in AD but increased in VaD and DLB.

partially addressed by the fact that meta-analyses combine data from multiple studies. The remaining criterion, whether the participation rate of eligible persons was at least 50%, was mostly unreported. Thus, it is unclear whether the samples were impacted by recruitment bias, potentially limiting the generalisability of the current findings.

Intra-rater reliability was very high, with an ICC of 0.90 (95% CI 0.78 - 0.96) and 99% agreement between the reporting quality scores assigned to the 11 criteria on two separate rating occasions (6 to 8 weeks apart).

3.3. Alzheimer's disease (AD)

Thirty-eight studies compared the levels of 62 different inflammatory markers (see Fig. 3: cells 1, 2, 4, 8, 9 & 12) in individuals with AD and controls. Fifty-two of these markers were only examined by one or two studies (36 were significant), with the remaining 10 markers being examined by 3 to 11 studies (3 were significant). Most studies ($N_{\text{studies}} = 27$) had fewer than 50 participants in the AD group, with the remaining 11 recruiting 100 to 200 participants in their dementia groups. Table 2 summarises the 39 inflammatory markers that differed significantly ($p < 0.05$) between people with and without AD (also shown in bold Fig. 3: cells 1, 4, & 9). The mean Hedges' g values ranged widely from a very large negative effect (Hedges' $g = -4.07$ for sIL2r, lower levels in AD) to a very large positive effect (Hedges' $g = 4.23$ for M-CSF1, higher levels in AD). The markers with the largest effect sizes ($g > 3.00$) were all assessed by single studies, with two having small to modest sample sizes (CD16⁺CD56⁺ $n = 43$; sIL2r $n = 20$), raising the possibility of unreliable results and a need for replication to confirm these findings. The remaining 23 markers (shown as non-bold and without the superscript 'a' in Fig. 3: cells 1, 2, 4, 8, 9, & 12) showed no difference between those with and without AD (see Online Resource D for the values for the non-significant inflammatory markers).

Of the 39 inflammatory markers that differed significantly between those with and without a diagnosis of AD, only three were examined by more than two studies (IL17A (IL17), IL1alpha, & IL10 were each examined by 4 studies). All three showed moderately higher levels in people with AD: IL17A (IL17) had a mean Hedges' g of 0.74 (95% CI 0.18 - 1.31, $p = 0.01$), IL1alpha had a mean Hedges' g of 0.57 (95% CI 0.04 - 1.09, $p = 0.035$), and IL10 showed a mean Hedges' g of 0.59 (95% CI 0.04 - 1.15, $p = 0.036$).

Between-study heterogeneity could only be assessed for these three inflammatory markers because the remainder were examined by too few studies. As seen in Table 3, the Q statistics for all three markers (IL17A, IL1alpha, IL10) indicate that there was significant heterogeneity in the Hedges' g values reported by individual studies. The I^2 statistics and 95% prediction interval values also demonstrate high between-study variability (see Table 3). Therefore, the findings of individual studies were inconsistent and not due to sampling error alone, instead indicating potential clinical or methodological (e.g. duration of dementia, participants' age, ApoE genotype) differences between studies.

Unfortunately, none of the markers that differed significantly between those with and without AD were examined by enough studies to permit a meta-regression to explore the potential contribution of moderator variables (e.g. age, sex) to the variability in study findings. Likewise, the potential impact of publication bias could not be assessed because there were too few studies.

The remaining 36 markers that differed significantly between persons with and without AD were assessed by only one or two studies each (see Table 2). Although there were more than 100 participants in the AD group for 10 of these markers (G-CSF, IL3, CCL15, CCL18, CCL5, CCL7, IgA, IL11, M-CSF1, myeloid dendritic cells) and more than 50 for a further 4 markers (GM-CSF, basophils, CCL2, IL1), replication is required to permit greater confidence in the findings. Caution is particularly indicated for the remaining 22 markers with small samples, especially those with 10 or fewer participants in the AD group (IL5, microglial activation anterior cingulate, microglial activation cortical,

Table 2

Inflammatory markers that differed significantly between persons with AD ($N_{\text{markers}} = 62$), VaD ($N_{\text{markers}} = 15$), DLB ($N_{\text{markers}} = 41$), or FTD ($N_{\text{markers}} = 13$) and their matched controls (ordered by number of studies, then alphabetically).

Inflammatory marker	N_{studies}	Effect size	95%CI LL	95%CI UL	p-value
AD					
IL17A (IL17)	4	0.74	0.18	1.31	0.010
IL1alpha	4	0.57	0.04	1.09	0.035
IL10	4	0.59	0.04	1.15	0.036
G-CSF	2	0.80	0.37	1.23	0.000
GM-CSF	2	0.44	0.05	0.83	0.028
IL3	2	1.39	1.05	1.73	0.000
basophils	1	-0.44	-0.74	-0.14	0.004
CCL15 (MIP1d)	1	-1.17	-1.40	-0.93	0.000
CCL18 (PARC)	1	2.85	2.54	3.16	0.000
CCL2 (MCP1)	1	0.75	0.45	1.05	0.000
CCL5 (RANTES)	1	0.65	0.42	0.87	0.000
CCL7 (MCP3)	1	1.13	0.89	1.36	0.000
CD16 ⁺ CD56 ⁺	1	-4.01	-4.78	-3.23	0.000
CD19 ⁺	1	1.01	0.54	1.49	0.000
CD3 ⁺ CD8 ⁺	1	0.65	0.19	1.11	0.005
CD4 ⁺ /CD8 ⁺ (ratio)	1	-2.90	-3.54	-2.26	0.000
CD42a	1	0.69	0.25	1.12	0.002
CD42a + CD14 ⁺	1	1.49	1.01	1.96	0.000
CD62P	1	0.99	0.55	1.44	0.000
IgA	1	0.25	0.04	0.45	0.017
IL1	1	0.77	0.26	1.28	0.003
IL11	1	-1.23	-1.47	-0.98	0.000
IL1Ra	1	0.54	0.04	1.04	0.033
IL5	1	1.77	0.79	2.75	0.000
INFalpha2	1	-1.51	-2.00	-1.02	0.000
M-CSF1	1	4.23	3.83	4.63	0.000
Microglial activation	1	1.37	0.33	2.41	0.010
Anterior cingulate					
Microglial activation	1	1.25	0.23	2.27	0.017
Cortical					
Microglial activation	1	1.33	0.29	2.36	0.012
Frontal lobe					
Microglial activation	1	1.79	0.68	2.91	0.002
Medial temporal lobe					
Microglial activation	1	1.89	0.76	3.03	0.001
Occipital lobe					
Microglial activation	1	1.54	0.47	2.61	0.005
Parietal lobe					
Microglial activation	1	1.44	0.39	2.49	0.007
Posterior cingulate					
Microglial activation	1	1.67	0.58	2.76	0.003
Striatum					
Microglial activation	1	1.38	0.33	2.42	0.010
Temporal lobe					
Microglial activation	1	1.58	0.50	2.65	0.004
Thalamus					
Myeloid dendritic cells	1	-3.86	-4.36	-3.36	0.000
sIL2r	1	-4.07	-5.14	-3.00	0.000
TRAIL	1	-0.56	-1.00	-0.12	0.013
VaD					
CCL18 (PARC)	1	-1.14	-1.64	-0.64	0.000
CCL5 (RANTES)	1	4.27	3.62	4.91	0.000
IL11	1	0.70	0.22	1.19	0.005
IL1alpha	1	-1.90	-2.12	-1.38	0.000
IL3	1	-2.08	-2.61	-1.56	0.000
IL34	1	-2.89	-3.36	-2.42	0.000
IL8 (CXCL8)	1	-0.96	-1.45	-0.46	0.000
M-CSF1	1	-0.94	-1.43	-0.45	0.000
TNF-alpha	1	-1.78	-2.30	-1.27	0.000
DLB					
IL8 (CXCL8)	2	-0.81	-1.33	-0.30	0.002
CCL15 (MIP1d)	1	4.76	4.19	5.34	0.000
CCL18 (PARC)	1	-0.68	-1.04	-0.32	0.000
CCL20 (MIP-3A)	1	0.90	0.29	1.51	0.004
CCL7 (MCP3)	1	-1.78	-2.17	-1.38	0.000
CD19+HLA-DR+	1	-0.53	-1.03	-0.03	0.039
G-CSF	1	-3.68	-4.18	-3.18	0.000
IL11	1	0.84	0.48	1.21	0.000
IL17A (IL17)	1	0.62	0.02	1.21	0.042
IL1alpha	1	-1.55	-1.93	-1.17	0.000

(continued on next page)

Table 2 (continued)

Inflammatory marker	N _{studies}	Effect size	95%CI LL	95%CI UL	p-value
IL2	1	0.79	0.18	1.39	0.011
IL3	1	−1.58	−1.97	−1.20	0.000
FTD					
CCL15 (MIP1d)	1	0.87	0.42	1.32	0.000
CCL18 (PARC)	1	−1.63	−2.10	−1.16	0.000
CCL5 (RANTES)	1	2.53	2.02	3.04	0.000
G-CSF	1	0.45	0.01	0.90	0.046
IL1alpha	1	−0.85	−1.30	−0.40	0.000
IL3	1	−0.84	−1.28	−0.39	0.000
IL6	1	1.31	0.95	1.67	0.000
IL8 (CXCL8)	1	0.88	0.43	1.33	0.000
M-CSF1	1	0.89	0.44	1.34	0.000

95% CI LL = 95% confidence interval lower limit. 95% CI UL = 95% confidence interval upper limit.

microglial activation frontal lobe, microglial activation medial temporal lobe, microglial activation occipital lobe, microglial activation parietal lobe, microglial activation posterior cingulate, microglial activation striatum, microglial activation temporal lobe, and microglial activation thalamus).

Of the 23 markers that did *not* differ between the AD and Control groups (see Online Resource D for details), only one – TNF-alpha – was examined by more than 10 studies (N_{studies} = 11), providing greater confidence that it did *not* differ between people with and without AD. A further six markers were examined by three to nine studies, also providing some confidence that these markers did *not* differ between people with and without AD: IL6 (N_{studies} = 9), IL1beta (N_{studies} = 8), IL8 (N_{studies} = 5), IgG (N_{studies} = 4), IFNgamma (N_{studies} = 3), and lymphocytes (N_{studies} = 3). However, the findings for the remaining 16 non-significant markers should be considered tentative because of limited data (N_{studies} = 1 or 2; see Fig. 3: cells 1, 2, 4, 8, 9 & 12).

Of note, where both peripheral and central measures of markers were collected by different studies (N_{markers} = 6), most showed no significant difference between persons with and without AD, regardless of whether the peripheral, central or combined measures were examined (IL6, IgG, IgM, and sTNF-R1). Two exceptions to this were IL1beta (N_{studies} = 8; 1 central measure) and TNF-alpha (N_{studies} = 11; 1 central measure). When the central measures of these markers were analysed separately, they were lower in people with AD, although only the IL1beta findings were significant (IL1beta: Hedges' $g = -1.77$, 95% CI -2.48 to -1.06, $p = 0.000$; TNF-alpha: Hedges' $g = -0.31$, 95% CI -0.91 to 0.30, $p = 0.322$). In contrast, no differences were found between people with and without AD when the peripheral measures of these markers were used (IL1beta: Hedges' $g = 0.45$, 95% CI 0.00 – 0.90, $p = 0.051$; TNF-alpha: Hedges' $g = 0.56$, 95% CI -0.29 – 1.41, $p = 0.194$).

3.4. Vascular Dementia (VaD)

Just two studies compared a total of 15 markers in people with vascular dementia (VaD) and their controls (see Fig. 3: cells 2, 3, & 9). Each marker was examined by a single study, with 84 participants in the VaD group for four markers (IL34, lymphocytes, neutrophils, white blood cells) and only 18 participants for the remaining 11 markers. Thus, the findings for VaD should be considered tentative. Table 2

summarises the results for the nine markers that differed significantly ($p < 0.05$) between people diagnosed with and without VaD, with Hedges' g values ranging from a very large negative effect ($g = -2.89$ for IL34; lower levels in those with VaD) to a very large positive effect ($g = 4.27$ for CCL5; higher levels in those with VaD). The single marker with a very large effect size ($g > 3.00$) was assessed by a single study examining a small sample (CCL5 $n = 18$); highlighting the need for replication to ensure the findings are robust. The remaining six markers did not differ significantly (see Online Resource D for details). Analyses to examine the mediating effects of potential covariates (age or sex) or publication bias were not possible due to the limited data.

3.5. Dementia with Lewy bodies (DLB)

Three studies compared 41 markers in persons with dementia with Lewy bodies (DLB) to those of healthy controls; all with modest sample sizes (N_{participants} = 19 to 37). Three markers (IL8, M-CSF1, TNF-alpha) were examined by two studies each (one was significant), with the rest examined by a single study. Table 2 provides summary data for the 12 inflammatory markers that differed significantly between people with and without DLB (see also Fig. 3: cells 4, 7 & 9), with the Hedges' g values ranging from a very large negative effect ($g = -3.68$ for G-CSF) to a very large positive effect ($g = 4.76$ for CCL15). The two markers with very large effect sizes ($g > 3.00$) were both assessed by the same single study with a modest sample size (CCL15 $n = 37$; G-CSF $n = 37$), underscoring the need for replication and confirmation of these findings. The remaining 29 markers did not differ significantly between people with DLB and their matched controls (see Fig. 3: cells 4, 7, 8, & 9 with specific data provided in Online Resource D). As was the case for VaD, the limited number of studies and participants restricts the interpretation of these results and prevented further analyses.

3.6. Fronto-temporal dementia (FTD)

Two studies examined 13 markers in persons with fronto-temporal dementia (FTD) and healthy controls (see Fig. 3: cells 8, 9, & 12). All markers, except one (TNF-alpha, N_{studies} = 2), were examined by single studies. There were just 22 participants in the FTD group for 10 of these markers, with higher participant numbers for the remaining three markers (N_{participants} = 92 for IL18 & IL6; N_{participants} = 114 for TNF-alpha). As shown in Table 2, the Hedges' g values for the nine inflammatory markers that differed significantly between those with and without FTD ranged widely from a very large negative effect ($g = -1.63$ for CCL18) to a very large positive effect ($g = 2.53$ for CCL5). The remaining 4 markers did not differ significantly between persons with and without FTD (see Online Resource D for details). Unfortunately, as was true for VaD and DLB, the limited participant and study numbers restricted the interpretation of these results and precluded additional analyses.

3.7. Comparison between AD, VaD, DLB and FTD

Finally, the results for each inflammatory marker were compared across the four dementia groups to explore whether there are similarities in the immune system response of persons with AD, VaD, DLB, or FTD. As seen in Fig. 3 (bold text), 46 of the 82 inflammatory markers

Table 3

Heterogeneity results for the inflammatory markers that differed significantly between people with AD and their healthy controls.

Alzheimer's Disease (AD)							
	Q	df(Q)	p	I ²	T ²	T	95% Prediction Interval
IL17A (IL17)	9.915	3	0.019	69.744	0.226	0.476	−1.651 - 3.134
IL1alpha	17.482	3	0.001	82.840	0.196	0.443	−1.661 - 2.791
IL10	7.879	3	0.049	61.925	0.191	0.437	−1.648 - 2.831

T = tau, T² = tau squared.

examined in this meta-analysis showed significant findings for one or more dementia groups (AD, VaD, DLB, FTD). Most ($N_{\text{markers}} = 28$) were examined in the context of one type of dementia, thus precluding comparison: 26 markers were examined in AD alone (see Fig. 3: cell 1), one in VaD (see Fig. 3: cell 3), and one in DLB (see Fig. 3: cell 7).

Of the remaining 18 markers that showed significant differences, 11 were examined (by different studies) across all four dementia groups (AD, VaD, DLB, FTD). However, none of these 11 markers showed consistent findings across dementia types, with only three markers (IL1alpha, IL3, CCL18) differing significantly in all four dementia groups (see Fig. 3: cell 9). Interestingly, as seen in Table 4, higher levels of IL1alpha, IL3 and CCL18 were seen in persons with AD, in contrast to lower levels in persons with VaD, DLB or FTD. The remaining eight markers that were examined across all four dementia types (see Fig. 3: cell 9) varied as to whether they were significant for one, two or three dementia groups and whether they were higher or lower in comparison to controls.

Just one marker (IL6) was examined in three dementia groups (AD, DLB and FTD; see Fig. 3: cell 8), but it only differed significantly from healthy controls in persons with FTD. The remaining six markers that differed significantly between those with and without dementia were examined in two dementia types (AD and DLB; see Fig. 3: cell 4), only one (IL17A) of which was significant for both dementias (see Table 4 for specific values).

Of the 36 inflammatory markers that did not differ between the dementia and control groups, 28 (see Table 4) were only examined for a single dementia type (see Fig. 3: cells 1, 3, & 7). The remaining eight non-significant markers were only examined in two dementia types: 2 in AD and VaD, 5 in AD and DLB, and 1 in AD and FTD (see Fig. 3: cells 2, 4, & 12, respectively).

Overall, these findings suggest minimal overlap in the immune cell and inflammatory mediator profiles associated with AD, VaD, DLB and FTD. However, these results should be considered tentative because most markers were examined in the context of a single type of dementia and, for many of the inflammatory markers that were examined across dementia types, the results were limited by the small number of studies and participants in the VaD ($N_{\text{studies}} = 2$; $N_{\text{participants}} = 18 - 84$), DLB ($N_{\text{studies}} = 3$, $N_{\text{participants}} = 19 - 37$) and FTD ($N_{\text{studies}} = 2$, $N_{\text{participants}} = 22 - 92$) groups.

4. Discussion

This study investigated the hypothesis that inflammation may contribute to the neurodegenerative processes that lead to dementia by examining the levels of inflammatory markers in people with dementia and healthy age- and sex-matched controls. It expanded upon the findings of previous meta-analyses, which examined inflammation in AD (Anuradha et al., 2022; Bradburn et al., 2019; Chen et al., 2018; Darweesh et al., 2018; Gautam et al., 2023; Hecheng et al., 2023; Holper et al., 2024; Jiang et al., 2018; Koyama et al., 2013; Lai et al., 2017; Liu et al., 2018; Mavroudis et al., 2021; Ng et al., 2018; Shen et al., 2019; Su et al., 2019; Swardfager et al., 2010; Zhou et al., 2023a) and all-cause/unspecified dementia (Darweesh et al., 2018; Gibson et al., 2021; Koyama et al., 2013; Pan et al., 2024). Importantly, unlike earlier meta-analyses, it excluded people with comorbid mental or physical health conditions with known inflammatory responses to determine whether some or all of the four most common types of dementia (AD/VaD/DLB/FTD) are independently associated with higher levels of inflammation. It also captured a broader range of potential inflammatory responses by including both immune cells (e.g. lymphocytes) and inflammatory mediators (e.g. cytokines).

Forty-one studies were eligible for inclusion; 38 examined AD, three examined DLB, and two each investigated VaD and FTD. Of the 82 inflammatory markers that were assessed, most ($N_{\text{markers}} = 56$) were examined in a single dementia subtype; predominantly AD ($N_{\text{markers}} = 36$). TNF-alpha was the most frequently assessed marker ($N_{\text{studies}} = 13$), followed by interleukin 6 (IL6) ($N_{\text{studies}} = 11$), with the remainder

examined by five or less studies. Most markers were examined by a single study ($N_{\text{markers}} = 59$) and the findings for these markers should be considered tentative, with greater weight placed on the findings supported by multiple studies.

People with AD demonstrated an inflammatory profile characterised by higher levels of six markers, which were assessed by multiple studies: IL17A (IL17) (mean Hedges' $g = 0.74$, $p = 0.010$), IL1alpha (mean Hedges' $g = 0.57$, $p = 0.035$), IL10 (mean $g = 0.59$, $p = 0.036$), G-CSF (mean $g = 0.80$, $p = 0.000$), GM-CSF (mean $g = 0.44$, $p = 0.028$), and IL3 (mean $g = 1.39$, $p = 0.000$). However, the findings for the latter three markers should be considered provisional because they were assessed by only two studies. An additional 33 markers differed in people with AD (24 markers showed higher levels and nine showed lower levels); however, these results require replication because they were each assessed by single studies and most ($N_{\text{markers}} = 22$) had fewer than 50 participants (see Table 2).

Of note, the higher levels of IL17A (IL17) ($N_{\text{studies}} = 4$) found in people with AD may be attributable to their higher levels of IL1alpha ($N_{\text{studies}} = 4$) and CCL18 ($N_{\text{studies}} = 1$) because these two markers can trigger an increase of IL17 (Cavalli et al., 2021; Kimura and Kishimoto, 2010; Mantovani et al., 2019; Singh et al., 2024). CCL18 can also elevate IL6 (Zeng et al., 2022). Intriguingly, there were no significant findings for elevated IL6 protein levels in AD ($N_{\text{studies}} = 9$); however, IL6 inflammatory activity, referring to the downstream biological effects of IL6, may still have been increased potentially via trans-signalling where IL6 binds to sIL6R and the complex (IL6/sIL6R) binds to gp130 cell surface receptor sites (Hampel et al., 1998; Vilaiyuk et al., 2019). The trend seen in the present study for lower levels of sIL6R ($N_{\text{studies}} = 2$) appears consistent with this, as with more of IL6/sIL6R bound to gp130 receptor sites on the cell surface, lower levels of these markers (IL6 and sIL6R) would be seen in the blood and CSF (Hampel et al., 1998; Vilaiyuk et al., 2019). This trans-signalling drives pro-inflammatory responses, as distinct from classical signalling (where IL6 binds to the IL6 cell-surface receptor), which is associated with anti-inflammatory responses (Nash et al., 2022; Scheller et al., 2011). Increased IL17A can disrupt the blood-brain barrier (Kebir et al., 2007), and elevated IL1alpha can induce an immune based activation of the endothelium – the single layer of cells lining the inside of all blood vessels and lymphatic vessels, which mediates immune transmigration responses (Feletou, 2011) – to attract immune cells across the blood-brain barrier (Cavalli et al., 2021; Varricchi et al., 2021; Williams, 2022). This raises the possibility of targeting immunomodulatory endothelial cells immune cell adhesion molecules and their ligands for new immunotherapeutic treatments (Amersfoort et al., 2022) in the fight against AD.

Heterogeneity analyses could only be performed for three markers (IL17A (IL17), IL1alpha, & IL10) that were investigated by 3 or more studies: all three showed significant variability in the findings from individual studies. This suggests that clinical and/or methodological differences between the studies – such as participants' age or sex – may have impacted on the findings. Unfortunately, it was not possible to investigate whether any such methodological differences contributed to the observed variability, or whether publication bias may have influenced the results, due to the small number of studies.

Different immune responses were seen in people with VaD, who showed higher levels of two (CCL5, IL11) out of the 15 markers that were examined and lower levels of seven other markers (CCL18, IL1alpha, IL3, IL34, IL8, M-CSF1, TNF-alpha). However, these findings should be considered provisional and would benefit from replication because each marker was only examined by a single study and, all but one (IL34 $N_{\text{participants}} = 84$), sampled fewer than 20 participants with dementia.

DLB was also associated with a distinct immune response that was defined by higher levels of five (CCL15, CCL20, IL11, IL17A, IL2) of the 41 inflammatory markers that were researched, and lower levels of an additional seven markers (IL8, CCL18, CCL7, CD19+HLA-DR+, G-CSF, IL1alpha, IL3). Like VaD, these findings are tentative because all but one marker (IL8 $N_{\text{studies}} = 2$, $N_{\text{participants}} = 56$) was examined by a single

Table 4

Comparison between findings from studies examining inflammatory markers in persons with AD, VaD, DLB, or FTD (and matched controls), rank-ordered by number of studies, then alphabetically.

Inflammatory marker	Total	AD		VaD		DLB		FTD	
	N _{studies}	g	N _{studies}	g	N _{studies}	g	N _{studies}	g	N _{studies}
Markers examined in AD, VaD, DLB and FTD									
TNFalpha	13	0.48	11	−1.78***	1	−0.71	2	−0.58	2
IL8 (CXCL8)	6	0.16	5	−0.96***	1	−0.81**	2	0.88***	1
IL1alpha	4	0.57*	4	−1.90***	1	−1.55***	1	−0.85***	1
G-CSF	2	0.80***	2	−0.21	1	−3.68***	1	0.45*	1
IL3	2	1.39***	2	−2.08***	1	−1.58***	1	−0.84***	1
M-CSF1	2	4.23***	1	−0.94***	1	0.02	2	0.89***	1
CCL15 (MIP-1d)	1	−1.17***	1	0.18	1	4.76***	1	0.87***	1
CCL18 (PARC)	1	2.85***	1	−1.14***	1	−0.68***	1	−1.63***	1
CCL5 (RANTES)	1	0.65***	1	4.27***	1	0.00	1	2.53***	1
CCL7 (MCP-3)	1	1.13***	1	0.17	1	−1.78***	1	0.00	1
IL11	1	−1.23***	1	0.70**	1	0.84***	1	0.42	1
Markers examined in AD, DLB and FTD									
IL6	11	−0.35	9		0	0.00	1	1.31***	1
Markers examined in AD, and VaD									
Lymphocytes	4	−0.70	3	−0.16	1				
White blood cells	2	0.01	1	0.14	1				
Markers examined in AD and DLB									
IL17A (IL17)	5	0.74*	4			0.62*	1		
IL10	5	0.59*	4			−0.05	1		
IFNgamma	4	0.30	3			−0.15	1		
GM-CSF	3	0.44*	2			−0.06	1		
IL12 (IL12p70)	3	0.36	2			0.35	1		
IL2	3	0.17	2			0.79*	1		
CCL2 (MCP-1)	2	0.75***	1			−0.15	1		
CD3 ⁺ CD4 ⁺	2	0.48	2			−0.45	1		
CD45RA + CCR7-	1	0.10	1			0.39	1		
CD45RA-CCR7+	1	0.11	1			0.24	1		
CD19+HLA-DR+	1	0.31	1			−0.53*	1		
Markers examined in AD and FTD									
IL18	3	0.20	2					−0.09	1
Markers examined only in AD									
IL1beta	8	0.20	8						
IgG	4	0.16	4						
IgM	2	0.08	2						
sIL6R	2	−1.18	2						
sTNF-R1 (sTNF-alpha_r)	2	1.28	2						
basophils	1	−0.44**	1						
CD14 ⁺	1	0.22	1						
CD16 + 56+	1	−4.01***	1						
CD19 ⁺	1	1.01***	1						
CD3 ⁺	1	−0.42	1						
CD4+/CD8+ (ratio)	1	−2.90***	1						
CD42a	1	0.69**	1						
CD42a + CD14 ⁺	1	1.49***	1						
CD62P	1	0.99***	1						
CD62P + CD14	1	−0.38	1						
CD3 ⁺ CD8 ⁺	1	0.65**	1						
IgA	1	0.25*	1						
IL1	1	0.77**	1						
IL1Ra	1	0.54*	1						
IL4	1	0.35	1						
IL5	1	1.77***	1						
INFalpha2	1	−1.51***	1						
Microglial activation Anterior Cingulate	1	1.37*	1						
Microglial activation Cortical	1	1.25*	1						
Microglial activation Frontal lobe	1	1.33*	1						
Microglial activation Hippocampus	1	0.98	1						
Microglial activation Medial Temporal lobe	1	1.79**	1						
Microglial activation Occipital lobe	1	1.89**	1						
Microglial activation Parietal lobe	1	1.54**	1						
Microglial activation Posterior Cingulate	1	1.44**	1						
Microglial activation Striatum	1	1.67**	1						
Microglial activation Temporal lobe	1	1.38*	1						
Microglial activation Thalamus	1	1.58**	1						
Myeloid dendritic cells	1	−3.86***	1						
sIL2r	1	−4.07***	1						
TRAIL	1	−0.56*	1						
Markers examined only in VaD									
IL34	1			−2.89***	1				
neutrophils	1			0.16	1				
Markers examined only in DLB									

(continued on next page)

Table 4 (continued)

Inflammatory marker	Total	AD		VaD		DLB		FTD	
	N _{studies}	g	N _{studies}	g	N _{studies}	g	N _{studies}	g	N _{studies}
CCL11 (eotaxin)	1					−0.15	1		
CCL12 (TARC)	1					−0.23	1		
CCL13 (MCP-4)	1					−0.37	1		
CCL20 (MIP-3A)	1					0.90**	1		
CCL26 (eotaxin 3)	1					−0.34	1		
CCL3 (MIP-1alpha)	1					−0.18	1		
CCL4 (MIP-1beta)	1					−0.18	1		
CXCL10 (IP10)	1					0.41	1		
IL13	1					−0.11	1		
IL15	1					−0.08	1		
IL16	1					−0.03	1		
IL22	1					0.41	1		
IL27	1					0.02	1		
IL7	1					0.17	1		
Macrophage derived chemokine (CCL22)	1					−0.35	1		
TNFB	1					0.10	1		
TNFR1 (CD120a)	1					0.39	1		
YKL40	1					0.56	1		

Notes: Inflammatory markers that differed significantly between the patient group and their controls are shown in **Bold**.

*p < 0.05. **p < 0.01. ***p < 0.001.

small-scale study, highlighting the need for further research.

A specific immune profile was also seen in people with FTD, who demonstrated *higher* levels of six (CCL15, CCL5, G-CSF, IL6, IL8, M-CSF1) of the 13 markers that were investigated and *lower* levels of a further three markers (CCL18, IL1alpha, IL3). Similar to VaD and DLB, these findings must be considered tentative and replication of these results is required because each marker was only assessed by a single study and, with one exception (IL6), recruited fewer than 25 participants with dementia.

Interestingly, people with AD showed *higher* levels of IL1alpha (N_{studies} = 4), IL3 (N_{studies} = 2) and CCL18 (N_{studies} = 1) whereas *lower* levels were seen in people with VaD, DLB and FTD. However, all three markers were assessed by single studies of VaD, DLB and FTD, limiting the confidence in these findings. Although higher levels of IL17A were seen in both AD (N_{studies} = 4) and DLB (N_{studies} = 1), these results likely reflect distinct immune processes, with higher IL1alpha and CCL18 levels potentially accountable for higher IL17A in AD, but not in DLB. Further delineating AD from the other three dementias, CCL5 was moderately elevated in AD (N_{studies} = 1), but was substantially higher in VaD (N_{studies} = 1) and FTD (N_{studies} = 1), and was not elevated in DLB (N_{studies} = 1). Of note, CCL5 has been implicated in atherosclerosis in clinical studies (Marquez et al., 2021) and disruption to the blood brain barrier in an animal study (Zhou et al., 2023b). Additionally, decreases of IL8 in VaD (N_{studies} = 1) and DLB (N_{studies} = 2) contrast with IL8 increases in FTD (N_{studies} = 1). Although these findings are tentative, they suggest that these three dementias (VaD/DLB/FTD) may have some overlapping immune mediated changes in common, but do not share identical inflammatory responses.

Further comparison of the inflammatory profiles associated with AD, VaD, DLB and FTD, which are otherwise clinically and pathologically distinct (Chin, 2023), was limited because less than one third (N_{markers} = 26) of the 82 inflammatory markers explored by this study were examined in more than one dementia subtype. Indeed, only 11 markers were assessed across all four dementia subtypes, with just one other assessed across three subtypes (AD, DLB & FTD), and a further 14 examined in relation to two subtypes (2 for AD & VaD, 11 for AD & DLB, 1 for AD & FTD). Of these 26 markers, there were 11 that demonstrated an inflammatory response in more than one type of dementia. However, they all showed divergent findings between the dementia subtypes, with some higher, some lower, and some not significantly different from the levels seen in healthy peers. Although provisional, these results suggest that AD, VaD, DLB, and FTD may each be associated with divergent immune responses and, therefore, potential therapeutic interventions

for these dementias might include targeting the specific profile of abnormal immune cells and mediators associated with each of the different dementia types.

To explore whether these immune changes in dementia relate to prior mental illness, as proposed by the ‘inflammatory mediated neurodegenerative’ hypothesis, inflammatory markers that were examined by multiple studies and differed between those with and without dementia were compared with levels reported in people with depression, anxiety, or PTSD from a recent meta-analysis designed to examine the same hypothesis (Kuring et al., 2023). The comparison showed that IL17A was elevated in people with AD and people with depression, and that G-CSF was elevated in people with AD, people with depression, and people with PTSD (see Table 4 above and Online Resource F from Kuring et al., 2023). This suggests that a history of depression or PTSD may be linked to the development of AD via certain inflammatory pathways. However, more studies are needed in other dementias to better understand the degree of any overlap in the inflammatory profiles of dementia and mental illness.

Unlike earlier meta-analyses that have examined inflammation in AD and/or all-cause/unspecified dementia (Anuradha et al., 2022; Bradburn et al., 2019; Chen et al., 2018; Darweesh et al., 2018; Gautam et al., 2023; Gibson et al., 2021; Hecheng et al., 2023; Holper et al., 2024; Jiang et al., 2018; Koyama et al., 2013; Lai et al., 2017; Liu et al., 2018; Mavroudis et al., 2021; Ng et al., 2018; Pan et al., 2024; Shen et al., 2019; Su et al., 2019; Swardfager et al., 2010; Zhou et al., 2023a), this study focussed on the four most common types of dementia (AD/VaD/DLB/FTD) in the *absence* of comorbid mental and physical health conditions that have a known inflammatory response. Despite having only a 51% overlap in the studies that were analysed here and in earlier meta-analyses, there are some similarities in the findings. Specifically, some previous meta-analyses have also reported *higher* levels of CCL15 (Hecheng et al., 2023; Zhou et al., 2023a), CCL2 (Hecheng et al., 2023; Zhou et al., 2023a), CCL5 (Hecheng et al., 2023), IL1alpha (Anuradha et al., 2022), IL10 (Shen et al., 2019) and IL17 (Gautam et al., 2023) in AD and *no difference* in the levels of TNFalpha (Anuradha et al., 2022; Chen et al., 2018; Ng et al., 2018; Swardfager et al., 2010), IL6 (Chen et al., 2018; Darweesh et al., 2018; Koyama et al., 2013; Ng et al., 2018), IL1beta (Anuradha et al., 2022; Chen et al., 2018), IL8 (Anuradha et al., 2022; Chen et al., 2018), and IFNgamma (Anuradha et al., 2022; Swardfager et al., 2010). There were also some differences. Importantly, IL3 and G-CSF were not examined by earlier meta-analyses and were all found to be elevated in people with AD. Conversely, five previous meta-analyses reported higher levels of IL6 (Anuradha et al., 2022; Lai

et al., 2017; Shen et al., 2019; Su et al., 2019; Swardfager et al., 2010) and IL1beta (Lai et al., 2017; Ng et al., 2018; Shen et al., 2019; Su et al., 2019; Swardfager et al., 2010), two reported higher levels of CXCL10 (IP10) (Hecheng et al., 2023; Zhou et al., 2023a), one reported higher levels of IFNgamma (Lai et al., 2017), and one reported higher levels of IL8 (Su et al., 2019). These latter findings contrast with the current study, which found no differences in these markers in persons diagnosed with AD. In fact, the three largest effect sizes from individual studies suggested lower levels of IL6 in people with AD, with an overall small (and non-significant) negative effect size in contrast to the positive effect size reported in earlier meta-analyses. Additionally, unlike our finding that there were higher levels of IL10 and GM-CSF in people with AD, two prior meta-analyses reported no differences in IL10 (Anuradha et al., 2022; Swardfager et al., 2010) and one reported no difference in GM-CSF (Anuradha et al., 2022). These disparities may reflect the more stringent study criteria that were adopted here, particularly in relation to the exclusion of participants with comorbid dementias, and mental and/or physical health conditions that are associated with inflammation. The more stringent inclusion criteria may also explain the discrepancies seen between the current meta-analysis and the only other meta-analysis to have compared marker levels obtained from blood and CSF samples. The earlier study reported no difference in the levels of IL6, TNF-alpha, and IL1beta in CSF, in contrast to higher levels of each of these markers in blood samples from people with AD compared with healthy peers (Swardfager et al., 2010). This is at odds with the current study, which found no difference in the levels of IL6 in CSF or blood samples, and lower levels of TNF-alpha and IL1beta in CSF samples from people with AD compared to their healthy peers but no difference in these markers when analysed in blood samples. Differences in some of the findings from peripheral and central measures of the same inflammatory markers serves to highlight the need for future studies to examine both PNS and CNS samples. This would assist in determining whether blood samples provide a truly reliable proxy measure of CNS inflammation in dementia. It would also provide a clearer understanding of whether any inflammatory changes that are observed reflect: (1) neuroinflammation, as suggested by the inflammatory-mediated neurodegeneration hypothesis, (2) more systemic peripheral inflammation, or (3) a combination of both.

The finding that AD and, possibly, VaD, DLB and FTD are each associated with inflammation offers some support for the hypothesis that dementia may result from inflammatory-mediated neurodegeneration. Interestingly, based on the available evidence, the nature of this immune response appears to differ across the dementia subtypes. Adding to the plausibility of the 'inflammatory-mediated neurodegeneration' hypothesis, IL17A was elevated in AD and, in an earlier study, was also elevated in people with depression (Kuring et al., 2023). Similarly, G-CSF was elevated in AD and was previously shown to be higher in people with depression and people with PTSD (Kuring et al., 2023). However, the fact that the existing data for VaD, DLB, FTD and most of the inflammatory markers is limited restricts our interpretation and confidence in the findings. Current interpretations are restricted both by the limited available data and by the observational and cross-sectional nature of the existing studies, which are not able to disentangle whether inflammation is a cause, consequence or epiphenomenon of dementia. The inflammatory-mediated neurodegeneration hypothesis requires further research and, more specifically, prospective longitudinal research – to examine whether mental illness triggers an inflammatory response that may later lead to the neurodegenerative changes that contribute to dementia. Specifically, future studies should explore whether: a) AD, VaD, DLB and FTD genuinely differ in their inflammatory profiles, as suggested here, which may indicate different pathological pathways; b) any of the inflammatory profiles seen in these dementia subtypes remain consistent across the different stages of dementia and are also seen in people with mental illnesses, as might be expected if mental illness-related inflammation contributes to dementia; c) the inflammation seen in people with mental illness occurred after

(rather than before) their symptom onset to clarify whether mental disorders are a trigger for, rather than a response to, inflammation; and d) people with mental illness who have an inflammatory response are at greater risk of later being diagnosed with dementia than individuals with mental illness who do not show these inflammatory changes. This research may aid in the identification of potential biomarkers for mental illness and dementia, and possibly the development of new or adjunct anti-inflammatory treatments for people with these conditions.

If depression, anxiety and/or PTSD are shown to be modifiable causal risk factors for dementia, then early and effective treatment of these mental illnesses should help to reduce the incidence of dementia, resulting in significant health and economic benefits. In contrast, a prodromal relationship between mental health and dementia should see an improvement in mental health symptoms with effective treatment, but no changes in the incidence of dementia and its associated quality-of-life and economic burden. Future research should therefore focus on whether mental illness is a causal risk factor for, rather than a prodromal symptom of, dementia. Repeated examination of inflammatory markers throughout adulthood (young adult/mid-life/older adult) in people with and without mental illness in prospective, longitudinal studies would aid our understanding of whether the risk of dementia differs according to the age at which these mental illnesses occur and whether inflammation mediates this risk.

The inflammatory-mediated neurodegenerative hypothesis posits that neuroinflammation, rather than peripheral inflammation, underpins the link between mental illness and dementia (Leonard, 2017). Most of the existing studies examined inflammatory markers from peripheral samples (e.g. blood), which may not be linked to neuroinflammation, though several inflammatory proteins can damage the blood brain barrier, and/or cross the intact blood-brain barrier. Future research should examine inflammatory markers using CNS measures (e.g. CSF samples or neuroimaging with PET tracers to measure microglia) to more directly examine neuroinflammation. If it is only peripheral markers that are associated with an increased risk of dementia, this would suggest that local neuroinflammation alone does not directly underpin the link between mental illness and dementia, instead implicating systemic pathological processes.

There are several limitations to this meta-analysis that must be acknowledged. First, nine studies were excluded because it was not possible to obtain essential data from the corresponding authors. However, all but two of these studies either reported data consistent with the current findings or reported on an immune marker not otherwise included in the present study. The remaining two studies (Dursun et al., 2015; Wang et al., 2014) reported results inconsistent with the current findings for IL1alpha and IL10 respectively, which, if included, may have resulted in non-significant findings for these two inflammatory markers, rather than the increase seen in people with AD reported here. There were insufficient studies to examine publication bias, so it is unclear whether these omissions impacted on the results. Second, dementia duration, ethnicity and socioeconomic status were not commonly reported, so their impact could not be examined, which may limit generalisability. Third, most of the 82 markers were only examined by one study for each of the dementia subtypes (AD/VaD/DLB/FTD), highlighting the tentative nature of these findings. Fourth, some inflammatory mediators in the blood and CSF do not arise only from immune cells because some are also produced by other non-immune cells, and whilst this would not alter the observed results, it may indicate a different underlying mechanism as to the levels observed. Fifth, it is possible that some participants may have had: (1) an undiagnosed physical or mental health condition that was missed when screening study participants, and/or (2) a prior mental health condition that triggered a chronic inflammatory response, and/or (3) another comorbid health condition that was not recognised as causing an inflammatory response; potentially confounding the current results. Sixth, dementia diagnoses were made using clinical criteria that did not necessarily include biological markers, so it is possible that some participants with mixed dementia

were unintentionally included, thus confounding the results. Finally, although just one reviewer assessed study quality, the intra-rater reliability analysis demonstrated highly consistent ratings.

In conclusion, AD, VaD, DLB and FTD each appear to display inflammatory changes that were not accounted for by comorbid mental or physical health conditions that are known to be associated with inflammation. Whether AD, VaD, DLB and FTD share immune cell and inflammatory mediator responses similar to those associated with mental illness requires additional examination. Further longitudinal research is also required to determine whether pathological inflammatory responses contribute to an increased risk of dementia in people with prior depression, anxiety and, possibly, PTSD.

CRedit authorship contribution statement

J.K. Kuring: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **J.L. Mathias:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **L. Ward:** Writing – review & editing, Supervision, Methodology, Conceptualization. **G. Tachas:** Writing – review & editing, Supervision, Methodology, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpsychires.2026.02.044>.

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